

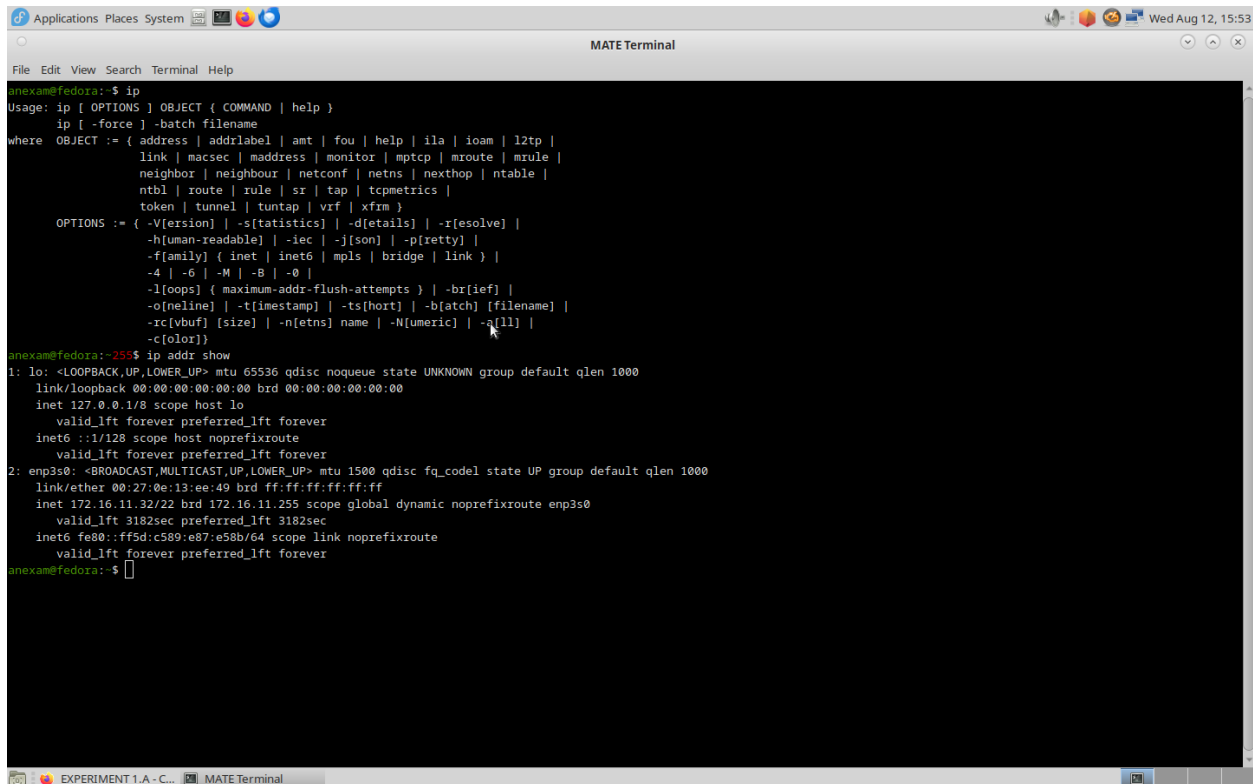
EXPERIMENT 1

Linux network commands

Aim:

To study and perform Linux Networking Commands

Screenshots:



```
anexam@fedora:~$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | amt | fou | help | ila | ioam | lltp |
                link | macsec | maddress | monitor | mptcp | mroute | mrule |
                neighbor | neighbour | netconf | netns | nexthop | ntable |
                ntbl | route | rule | sr | tap | tcpmetrics |
                token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                  -h[uman-readable] | -i[ec] | -j[son] | -p[retty] |
                  -f[amily] { inet | inet6 | mpls | bridge | link } |
                  -4 | -6 | -M | -B | -0 |
                  -l[oops] { maximum-addr-flush-attempts } | -b[r[ief] |
                  -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                  -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                  -c[olor]}
anexam@fedora:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:27:0e:13:ee:49 brd ff:ff:ff:ff:ff:ff
    inet 172.16.11.32/22 brd 172.16.11.255 scope global dynamic noprefixroute enp3s0
        valid_lft 3182sec preferred_lft 3182sec
    inet6 fe80::ff5d:c589:e87:e58b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
anexam@fedora:~$
```

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link/ether 00:27:0e:13:ee:49 brd ff:ff:ff:ff:ff:ff
inet 172.16.11.32/22 brd 172.16.11.255 scope global dynamic noprefixroute enp3s0
    valid_lft 3182sec preferred_lft 3182sec
inet6 fe80::ff5d:c589:e87:e58b/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
anexam@fedora:~$ ifconfig enp3s0
enp3s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.11.32 netmask 255.255.252.0 broadcast 172.16.11.255
    inet6 fe80::ff5d:c589:e87:e58b prefixlen 64 scopeid 0x20<link>
    ether 00:27:0e:13:ee:49 txqueuelen 1000 (Ethernet)
    RX packets 41102 bytes 31956744 (30.4 MiB)
    RX errors 0 dropped 34 overruns 0 frame 0
    TX packets 11027 bytes 2306068 (2.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

anexam@fedora:~$ ping google.com
PING google.com (142.250.77.142): 56(84) bytes of data:
64 bytes from maa0516-in-f14.1e100.net (142.250.77.142): icmp_seq=1 ttl=118 time=11.7 ms
64 bytes from maa0516-in-f14.1e100.net (142.250.77.142): icmp_seq=2 ttl=118 time=5.09 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=3 ttl=118 time=6.40 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=4 ttl=118 time=3.85 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=5 ttl=118 time=3.68 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=6 ttl=118 time=2.91 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=7 ttl=118 time=3.36 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=8 ttl=118 time=15.9 ms
64 bytes from maa0516-in-f14.1e100.net (142.250.77.142): icmp_seq=9 ttl=118 time=5.15 ms
64 bytes from maa0516-in-f14.1e100.net (142.250.77.142): icmp_seq=10 ttl=118 time=5.22 ms
64 bytes from maa0516-in-f14.1e100.net (142.250.77.142): icmp_seq=11 ttl=118 time=3.65 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=12 ttl=118 time=3.85 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=13 ttl=118 time=11.0 ms
64 bytes from maa0516-in-f14.1e100.net (142.250.77.142): icmp_seq=14 ttl=118 time=3.88 ms
64 bytes from maa0516-in-f14.1e100.net (142.250.77.142): icmp_seq=15 ttl=118 time=4.10 ms
64 bytes from lcaaaa-au-in-f14.1e100.net (142.250.77.142): icmp_seq=16 ttl=118 time=3.00 ms
^C
--- google.com ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15023ms
rtt min/avg/max/mdev = 2.913/5.801/15.911/3.637 ms
anexam@fedora:~$ traceroute google.com
traceroute to google.com (142.250.77.142), 30 hops max, 60 byte packets
 1 _gateway (172.16.8.1) 0.153 ms 0.179 ms 0.164 ms
 2 61.95.128.185 (61.95.128.185) 12.572 ms 7.908 ms 3.729 ms
 3 116.119.61.126 (116.119.61.126) 12.493 ms 7.908 ms 18.275 ms
 4 * 116.119.68.188 (116.119.68.188) 3.146 ms 116.119.161.149 (116.119.161.149) 12.235 ms
 5 * 142.250.169.206 (142.250.169.206) 7.534 ms 72.14.197.10 (72.14.197.10) 3.804 ms
 6 * 64.233.174.17 (64.233.174.17) 11.143 ms *
 7 142.251.55.228 (142.251.55.228) 3.276 ms 142.251.55.232 (142.251.55.232) 19.317 ms 142.251.55.88 (142.251.55.88) 3.020 ms
 8 * 142.251.55.63 (142.251.55.63) 7.454 ms 172.253.75.14 (172.253.75.14) 9.388 ms
 9 maa0516-in-f14.1e100.net (142.250.77.142) 8.124 ms 14.960 ms 4.053 ms
anexam@fedora:~$
```

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9 maa0516-in-f14.1e100.net (142.250.77.142) 8.124 ms 14.960 ms 4.053 ms
anexam@fedora:~$ tracepath google.com
17: [LOCALHOST] pmtu 1500
 1: _gateway 0.410ms
 1: _gateway 0.425ms
 2: 61.95.128.185 6.419ms
 3: 116.119.61.126 12.586ms asym 5
 4: 182.79.198.0 3.227ms
 5: no reply
 6: no reply
 7: no reply
 8: no reply
 9: no reply
10: no reply
11: no reply
12: no reply
13: no reply
14: no reply
15: no reply
16: no reply
17: no reply
18: no reply
19: no reply
20: no reply
^C
anexam@fedora:~$ ss -tlns
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 0.0.0.0:22 0.0.0.0:* LISTEN
tcp 0 0 0.0.0.0:53 0.0.0.0:* LISTEN
tcp 0 0 0.0.0.0:80 0.0.0.0:* LISTEN
tcp 0 0 0.0.0.0:443 0.0.0.0:* LISTEN
tcp 0 0 172.16.11.32:49696 142.251.220.113:443 TIME_WAIT
tcp 0 0 172.16.11.32:54092 34.54.185.247:443 TIME_WAIT
tcp 0 0 172.16.11.32:57108 142.251.43.42:443 TIME_WAIT
tcp 0 0 172.16.11.32:43564 142.251.43.202:443 TIME_WAIT
tcp 0 0 172.16.11.32:33542 142.250.286.3:443 TIME_WAIT
tcp 0 0 172.16.11.32:50868 34.187.243.93:443 ESTABLISHED
tcp 0 0 172.16.11.32:46710 142.251.43.226:443 TIME_WAIT
tcp 0 0 172.16.11.32:44138 142.251.221.97:443 TIME_WAIT
tcp 0 0 172.16.11.32:37790 142.250.77.106:443 TIME_WAIT
tcp 0 0 172.16.11.32:44722 172.217.119.4:443 TIME_WAIT
tcp 0 0 172.16.11.32:37774 142.250.77.106:443 TIME_WAIT
tcp6 0 0 :::1631 :::* LISTEN
tcp6 0 0 :::5355 :::* LISTEN
udp 0 0 0.0.0.0:54:53 0.0.0.0:*
udp 0 0 0.0.0.0:53:53 0.0.0.0:*
udp 0 0 172.16.11.32:68 172.16.8.1:67 ESTABLISHED
udp 0 0 0.0.0.0:1:323 0.0.0.0:*
```

```
Applications Places System MATE Terminal Wed Aug 12, 16:06
MATE Terminal
File Edit View Search Terminal Help
anexam@fedora:~$ netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 0.0.0.0:54:53 0.0.0.0:* LISTEN
tcp 0 0 0.0.0.0:5355 0.0.0.0:* LISTEN
tcp 0 0 0.0.0.0:1:631 0.0.0.0:* LISTEN
tcp 0 0 0.0.0.0:53:53 0.0.0.0:* LISTEN
tcp 0 0 0.0.0.0:53:53 0.0.0.0:* LISTEN
tcp 0 0 172.16.11.32:49696 142.251.220.113:443 TIME_WAIT
tcp 0 0 172.16.11.32:54092 34.54.185.247:443 TIME_WAIT
tcp 0 0 172.16.11.32:57108 142.251.43.42:443 TIME_WAIT
tcp 0 0 172.16.11.32:43564 142.251.43.202:443 TIME_WAIT
tcp 0 0 172.16.11.32:33542 142.250.206.3:443 TIME_WAIT
tcp 0 0 172.16.11.32:50368 34.107.249.93:443 ESTABLISHED
tcp 0 0 172.16.11.32:46710 142.251.43.226:443 TIME_WAIT
tcp 0 0 172.16.11.32:44138 142.251.221.97:443 TIME_WAIT
tcp 0 0 172.16.11.32:37790 142.250.77.106:443 TIME_WAIT
tcp 0 0 172.16.11.32:44722 172.217.119.4:443 TIME_WAIT
tcp 0 0 172.16.11.32:37774 142.250.77.106:443 TIME_WAIT
tcp6 0 0 :::1:631 :::* LISTEN
tcp6 0 0 :::5355 :::* LISTEN
udp 0 0 0.0.0.0:54:53 0.0.0.0:*
udp 0 0 0.0.0.0:53:53 0.0.0.0:*
udp 0 0 172.16.11.32:68 172.16.8.1:67 ESTABLISHED
udp 0 0 0.0.0.0:1:323 0.0.0.0:*
udp 0 0 0.0.0.0:45674 0.0.0.0:*
udp 0 0 0.0.0.0:5353 0.0.0.0:*
udp 0 0 0.0.0.0:5355 0.0.0.0:*
udp 0 0 0.0.0.0:42233 0.0.0.0:*
udp 0 0 0.0.0.0:40454 0.0.0.0:*
udp6 0 0 :::38920 :::*
udp6 0 0 :::1:323 :::*
udp6 0 0 :::5353 :::*
udp6 0 0 :::5355 :::*
raw6 0 0 :::58 :::* 7

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags Type State I-Node Path
unix 3 [ ] STREAM CONNECTED 27255
unix 3 [ ] STREAM CONNECTED 24136
unix 3 [ ] STREAM CONNECTED 27847
unix 3 [ ] STREAM CONNECTED 26786 @/tmp/.X11-unix/X0
unix 3 [ ] STREAM CONNECTED 25717 @/tmp/.X11-unix/X0
unix 3 [ ] STREAM CONNECTED 25296
unix 3 [ ] SEQPACKET CONNECTED 26338
unix 3 [ ] STREAM CONNECTED 25148 /run/user/1001/bus
unix 3 [ ] STREAM CONNECTED 20377 /run/dbus/system_bus_socket
unix 2 [ ] DGRAM CONNECTED 21924
unix 2 [ ] DGRAM CONNECTED 17282
unix 3 [ ] SEQPACKET CONNECTED 36833
unix 3 [ ] STREAM CONNECTED 25143
```

```
Applications Places System MATE Terminal Wed Aug 12, 16:16
MATE Terminal
File Edit View Search Terminal Help
anexam@fedora:~$ ss -tuln
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:45674 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:5353 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:5355 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:42233 0.0.0.0:*
udp UNCONN 0 0 0.0.0.0:40454 0.0.0.0:*
udp UNCONN 0 0 [::]:38920 [::]:*
udp UNCONN 0 0 [::]:323 [::]:*
udp UNCONN 0 0 [::]:5353 [::]:*
udp UNCONN 0 0 [::]:5355 [::]:*
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
tcp LISTEN 0 4096 0.0.0.0:5355 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.1:631 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
tcp LISTEN 0 4096 [::]:631 [::]:*
tcp LISTEN 0 4096 [::]:5355 [::]:*

anexam@fedora:~$ hostname
fedora
anexam@fedora:~$ nslookup google.com
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: google.com
Address: 142.251.43.78
Name: google.com
Address: 2404:6800:4007:817::200e

anexam@fedora:~$
```

```
Applications Places System MATE Terminal
Name: google.com
Address: 2404:6800:4007:817::200e

anexam@fedora:~$ dig google.com

; <<>> DiG 9.18.17 <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<<- opcode: QUERY, status: NOERROR, id: 29181
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A
;; ANSWER SECTION:
google.com.                192     IN      A      142.251.43.78

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Aug 12 16:17:27 IST 2026
;; MSG SIZE rcvd: 55

anexam@fedora:~$ host google.com
google.com has address 142.251.43.78
google.com has IPv6 address 2404:6800:4007:817::200e
google.com mail is handled by 10 smtp.google.com.
anexam@fedora:~$ arp -a
_gateway (172.16.8.1) at 7c:5a:1c:cf:be:45 [ether] on enp3s0
anexam@fedora:~$ ip neigh show
172.16.8.1 dev enp3s0 lladdr 7c:5a:1c:cf:be:45 REACHABLE
anexam@fedora:~$ route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 172.16.8.1 0.0.0.0 UG 100 0 0 enp3s0
172.16.8.0 0.0.0.0 255.255.252.0 U 100 0 0 enp3s0
anexam@fedora:~$ ip route show
default via 172.16.8.1 dev enp3s0 proto dhcp src 172.16.11.32 metric 100
172.16.8.0/22 dev enp3s0 proto kernel scope link src 172.16.11.32 metric 100
anexam@fedora:~$
```

```
Applications Places System MATE Terminal
anexam@fedora:~$ curl https://google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://www.google.com/">here</A>.
</BODY></HTML>
anexam@fedora:~$ wget https://example.com/file.zip
--2026-08-12 16:30:25-- https://example.com/file.zip
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:90c5:f2c5:0:ef6b:ff98
Connecting to example.com (example.com)[172.66.147.243]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-08-12 16:30:25 ERROR 404: Not Found.

anexam@fedora:~$ scp ca.py student@host:/home/student
ssh: Could not resolve hostname host: Name or service not known
scp: Connection closed
anexam@fedora:~$ ftp ftp.example.com
bash: ftp: command not found
anexam@fedora:~$ ftp www.google.com
bash: ftp: command not found
anexam@fedora:~$ telnet 192.168.1.10 80
bash: telnet: command not found
anexam@fedora:~$ telnet 192.168.1.10:80
bash: telnet: command not found
anexam@fedora:~$ nc -zv 192.168.1.10 22
Ncat: Version 7.95 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
anexam@fedora:~$ nmap 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-12 16:33 IST
Nmap scan report for 192.168.1.1
Host is up (0.0023s latency).
Not shown: 977 filtered tcp ports (no-response), 20 filtered tcp ports (net-unreach)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 4.71 seconds
anexam@fedora:~$
```

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Applications Places System | MATE Terminal | Wed Aug 12, 16:38
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anexam@fedora:~$ nc -zv 192.168.1.10 22
Ncat: Version 7.95 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
anexam@fedora:~$ nmap 192.168.1.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-12 16:33 IST
Nmap scan report for 192.168.1.1
Host is up (0.0023s latency).
Not shown: 977 filtered tcp ports (no-response), 20 filtered tcp ports (net-unreach)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 4.71 seconds
anexam@fedora:~$ sudo tcpdump -i eth0
[sudo] password for anexam:
anexam is not in the sudoers file.
anexam@fedora:~$ ethtool eth0
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: Operation not permitted
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
No data available
anexam@fedora:~$ iwconfig
bash: iwconfig: command not found
anexam@fedora:~$ nmcli device status
DEVICE  TYPE      STATE      CONNECTION
enp3s0  ethernet  connected  Wired connection 1
lo      loopback  connected (externally)  lo
anexam@fedora:~$ sudo systemctl restart NetworkManager
[sudo] password for anexam:
anexam is not in the sudoers file.
```

RESULT:

Thus the Linux network commands was studied and executed successfully.